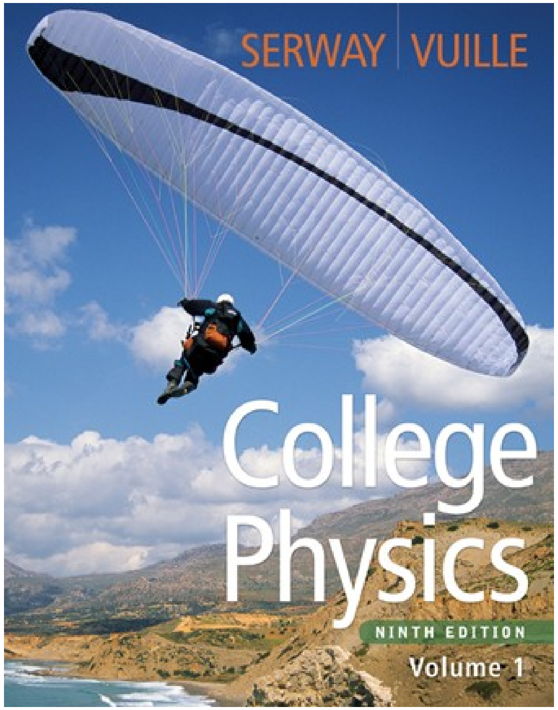

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Chapter Ten

Thermal Physics

Thermal Physics

- Thermal physics is the study of
  - Temperature
  - Heat
  - How these affect matter

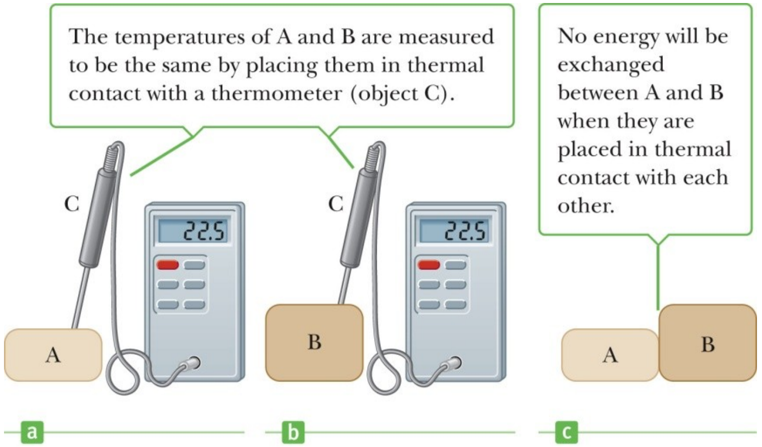
Thermal Physics, cont

- Descriptions require definitions of temperature, heat and internal energy
- Heat leads to changes in internal energy and therefore to changes in temperature
- Gases are critical in harnessing internal energy to do work

Definitions

- The process by which energy is exchanged between objects because of temperature differences is called *heat*
- Objects are in *thermal contact* if energy can be exchanged between them
- *Thermal equilibrium* exists when two objects are in thermal contact with each other and there is no net exchange of energy between them

Zeroth Law of Thermodynamics



- If objects A and B are separately in thermal equilibrium with a third object, C, then A and B are in thermal equilibrium with each other
  - Object C could be the thermometer
- Allows a definition of temperature

Temperature from the Zeroth Law

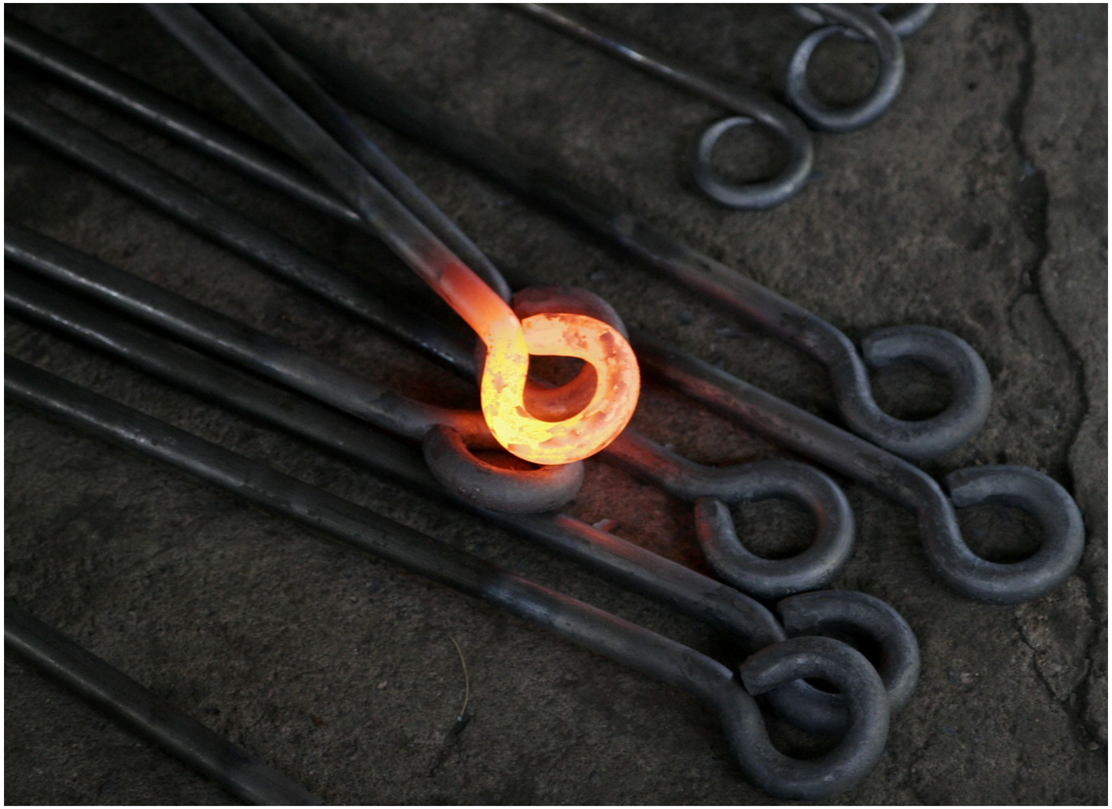
- *Temperature* is the property that determines whether or not an object is in thermal equilibrium with other objects
- Two objects in thermal equilibrium with each other are at the *same temperature*

Thermometers

- Used to measure the temperature of an object or a system
- Make use of physical properties that change with temperature
- Many physical properties can be used
  - Volume of a liquid
  - Length of a solid
  - Pressure of a gas held at constant volume
  - Volume of a gas held at constant pressure
  - Electric resistance of a conductor
  - Color of a very hot object

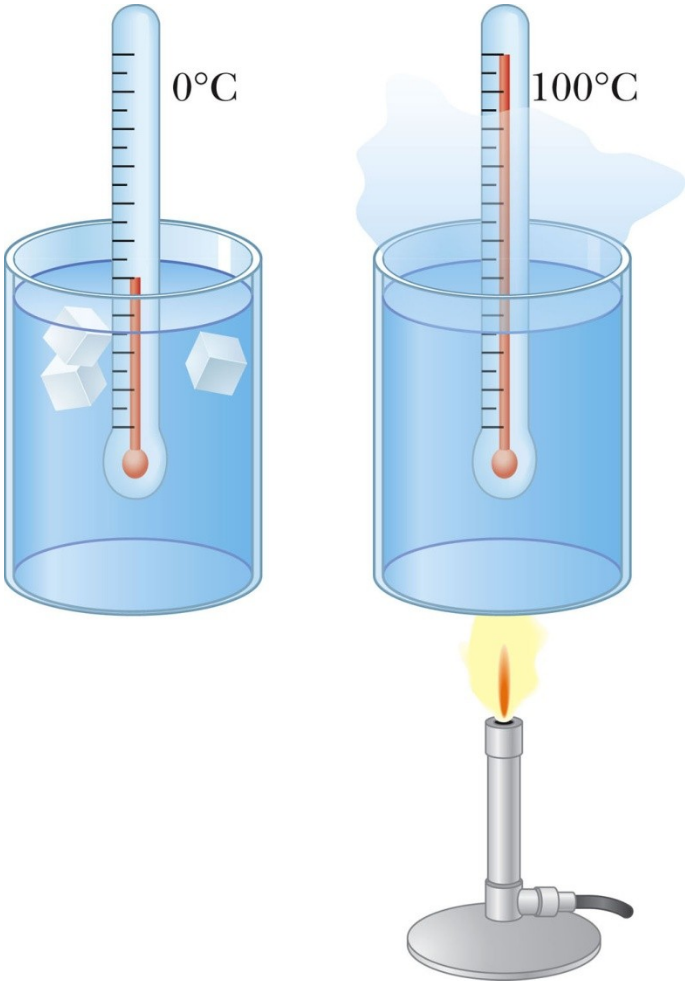
Thermometers

– Color of a very hot object



Thermometers, cont

- A mercury thermometer is an example of a common thermometer
- The level of the mercury rises due to thermal expansion
- Temperature can be defined by the height of the mercury column



Temperature Scales

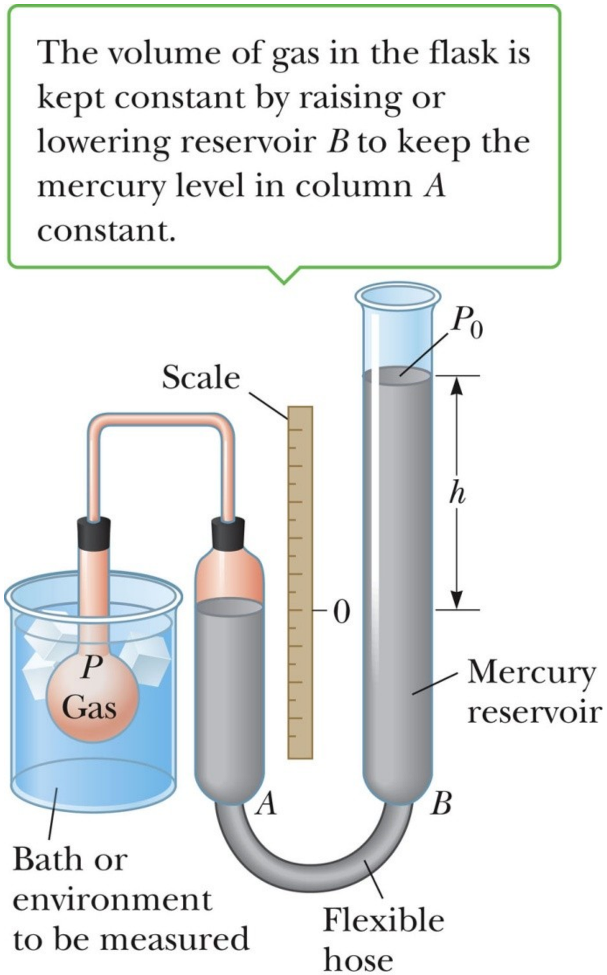
- Thermometers can be calibrated by placing them in thermal contact with an environment that remains at constant temperature
  - Environment could be mixture of ice and water in thermal equilibrium
  - Also commonly used is water and steam in thermal equilibrium

Celsius Scale

- Temperature of an ice-water mixture is defined as 0° C
  - This is the *ice point* or the *freezing point* of water
- Temperature of a water-steam mixture is defined as 100° C
  - This is the *steam point* or the *boiling point* of water
- Distance between these points is divided into 100 segments or degrees

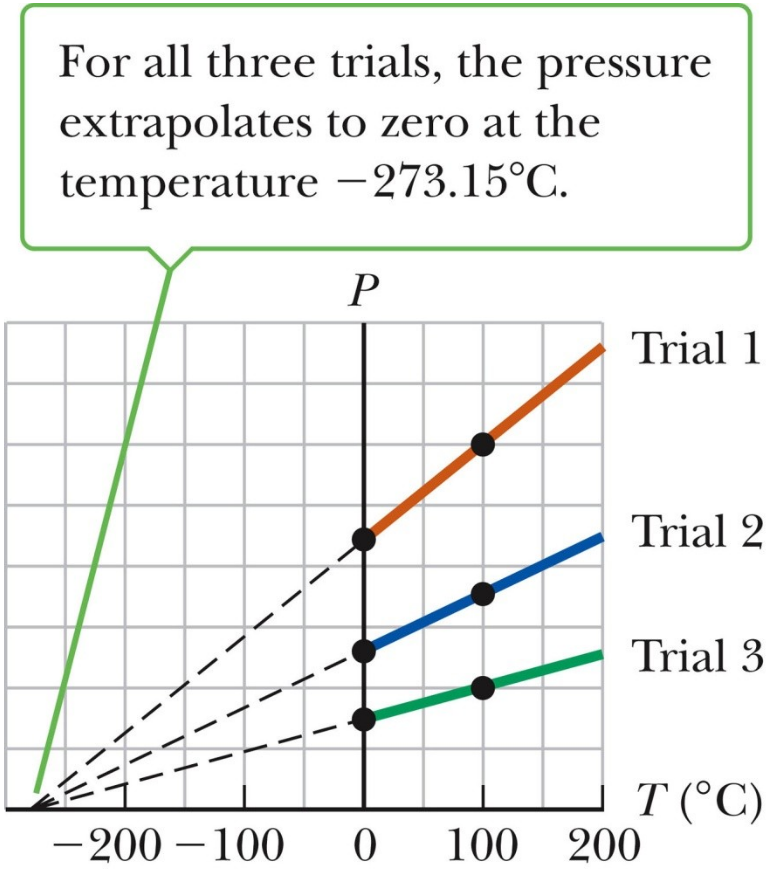
Gas Thermometer

- Temperature readings are nearly independent of the gas
  - As long as the gas pressure is low
  - The temperature needs to be well above the temperature at which the gas liquefies
- Pressure varies with temperature when maintaining a constant volume



Pressure-Temperature Graph

- All gases extrapolate to the same temperature at zero pressure
- This temperature is *absolute zero*



Kelvin Scale

- When the pressure of a gas goes to zero, its temperature is −273.15° C
  - This temperature is called *absolute zero*
- This is the zero point of the Kelvin scale
  - −273.15° C = 0 K
- To convert: $T_C = T - 273.15$
  - The size of the degree in the Kelvin scale is the same as the size of a Celsius degree

Modern Definition of Kelvin Scale

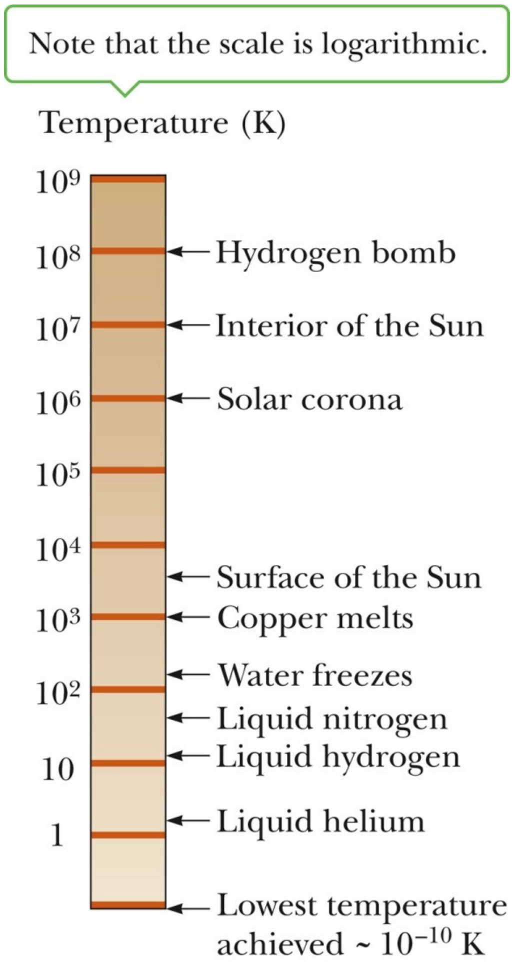
- Defined in terms of two points
  - Agreed upon by International Committee on Weights and Measures in 1954
- First point is absolute zero
- Second point is the *triple point* of water
  - The triple point is the single point where water can exist as solid, liquid, and gas in equilibrium
  - Single temperature and pressure
  - Occurs at 0.01° C and P = 4.58 mm Hg

Modern Definition of Kelvin Scale, cont

- The temperature of the triple point of water on the Kelvin scale is 273.16 K
- Therefore, the current definition of the Kelvin is defined as 1/273.16 of the temperature of the triple point of water

Some Kelvin Temperatures

- Some representative Kelvin temperatures
- Note, this scale is logarithmic
- Absolute zero has never been reached



Fahrenheit Scales

- Most common temperature scale used in the US
- Temperature of the ice point is 32° F
- Temperature of the steam point is 212° F
- 180 divisions between the points

Converting Among Temperature Scales

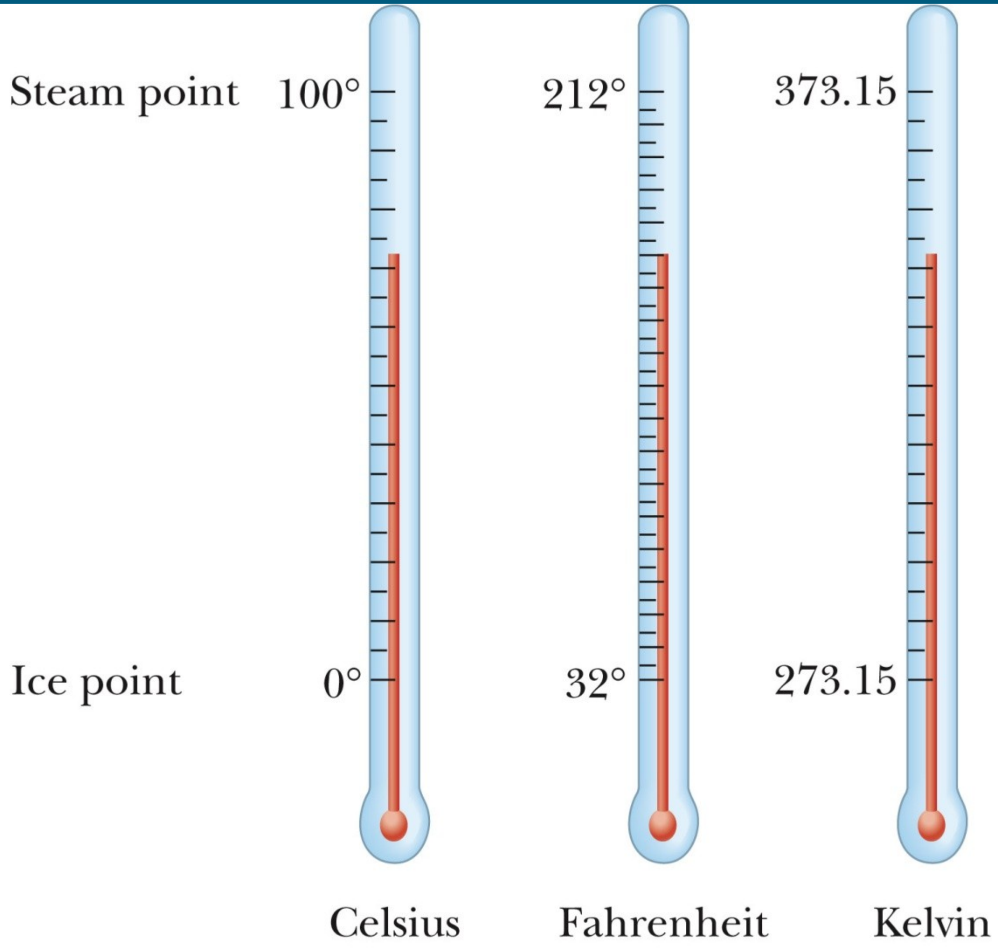
$$T_C = T_K - 273.15$$

$$T_F = \frac{9}{5}T_C + 32$$

$$T_C = \frac{5}{9}\left(T_F - 32\right)$$

$$\Delta T_F = \frac{9}{5}\Delta T_C$$

Comparing Temperature Scales



Example

Convert 50°C into °F and K.

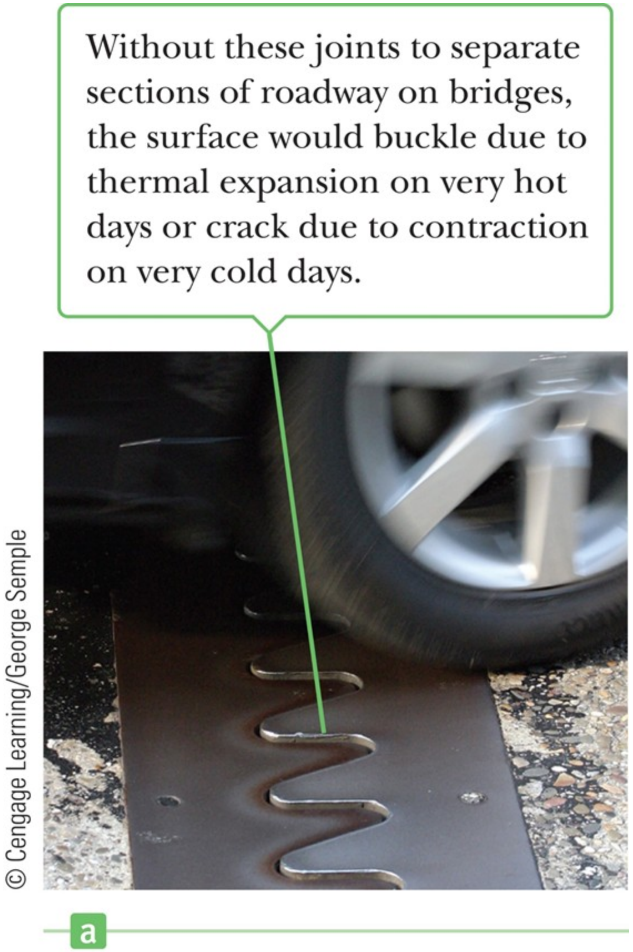
Example

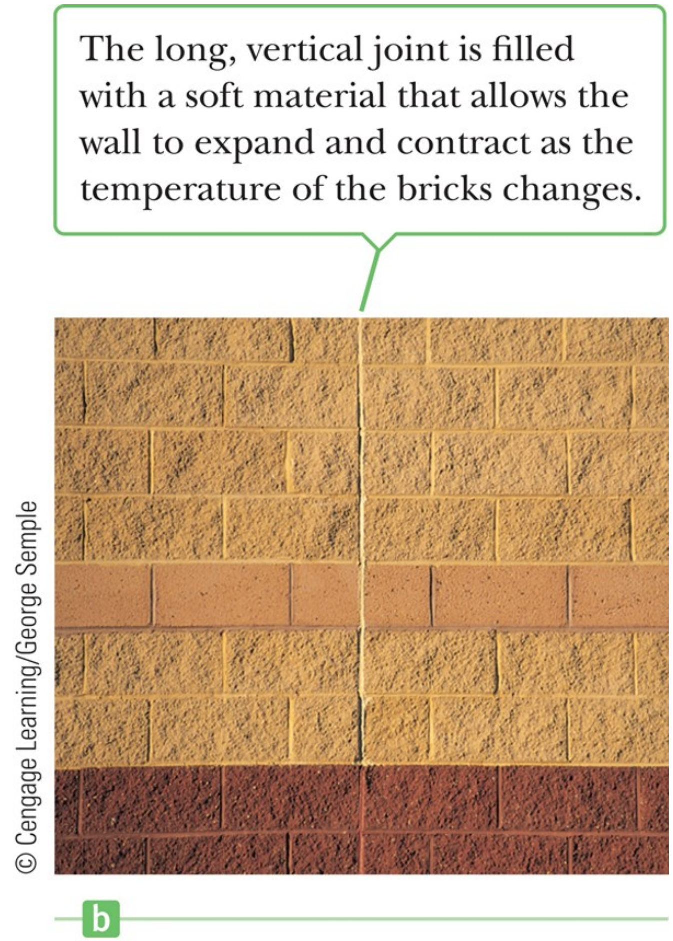
An air conditioning system reduces the temperature of a house from 90°F to 70°F. How much is the change in temperature in F, C and K?

Thermal Expansion

- The thermal expansion of an object is a consequence of the change in the average separation between its constituent atoms or molecules
- At ordinary temperatures, molecules vibrate with a small amplitude
- As temperature increases, the amplitude increases
  - This causes the object as a whole to expand

Thermal Expansion of Solids and Liquids (1 of 7)





Linear Expansion

- For small changes in temperature

$$\Delta L = \alpha\, L_o\, \Delta T \text{ or } L - L_o = \alpha\, L_o \left(T - T_o\right)$$

- α, the coefficient of linear expansion, depends on the material
  - See table
  - These are average coefficients, they can vary somewhat with temperature

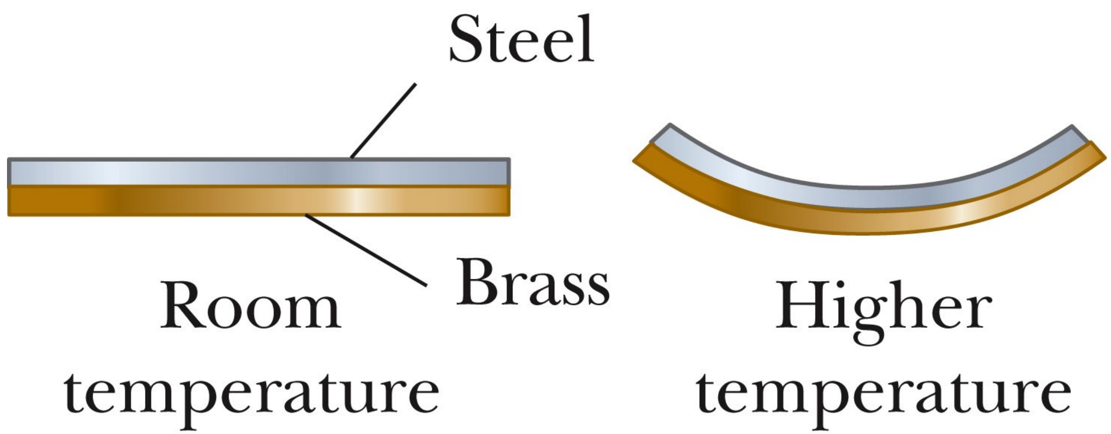
Some Coefficients

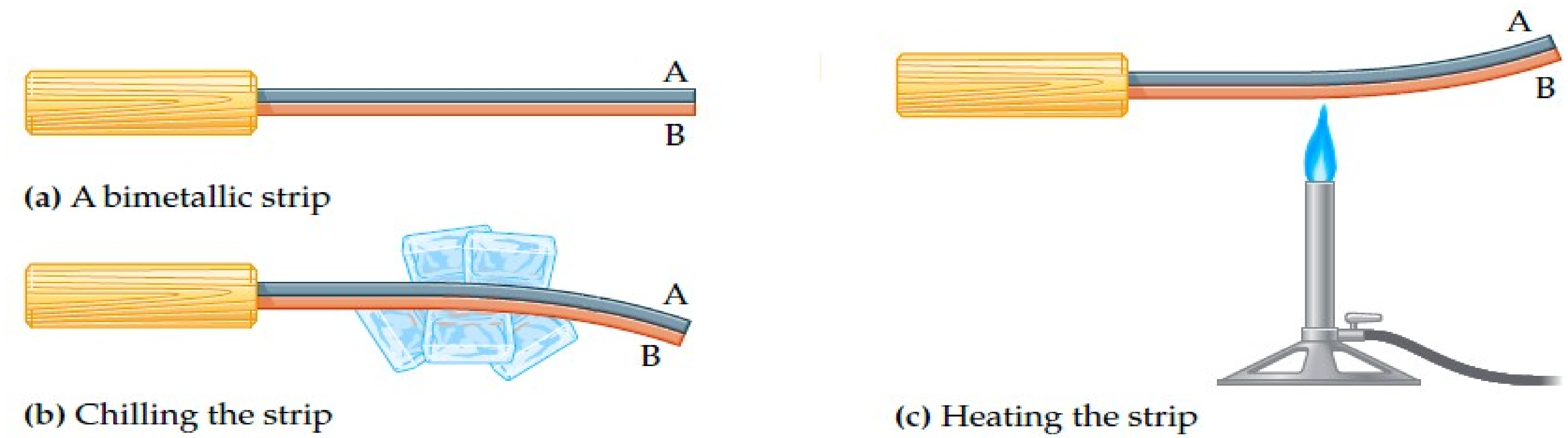
TABLE 19.1 *Average Expansion Coefficients for Some Materials Near Room Temperature*

Material (Solids)	Average Linear Expansion Coefficient $(\alpha)(°C)^{-1}$	Material (Liquids and Gases)	Average Volume Expansion Coefficient $(\beta)(°C)^{-1}$
Aluminum	24×10^{-6}	Acetone	1.5×10^{-4}
Brass and bronze	19×10^{-6}	Alcohol, ethyl	1.12×10^{-4}
Concrete	12×10^{-6}	Benzene	1.24×10^{-4}
Copper	17×10^{-6}	Gasoline	9.6×10^{-4}
Glass (ordinary)	9×10^{-6}	Glycerin	4.85×10^{-4}
Glass (Pyrex)	3.2×10^{-6}	Mercury	1.82×10^{-4}
Invar (Ni–Fe alloy)	0.9×10^{-6}	Turpentine	9.0×10^{-4}
Lead	29×10^{-6}	Air[a] at 0°C	3.67×10^{-3}
Steel	11×10^{-6}	Helium[a]	3.665×10^{-3}

[a]Gases do not have a specific value for the volume expansion coefficient because the amount of expansion depends on the type of process through which the gas is taken. The values given here assume the gas undergoes an expansion at constant pressure.

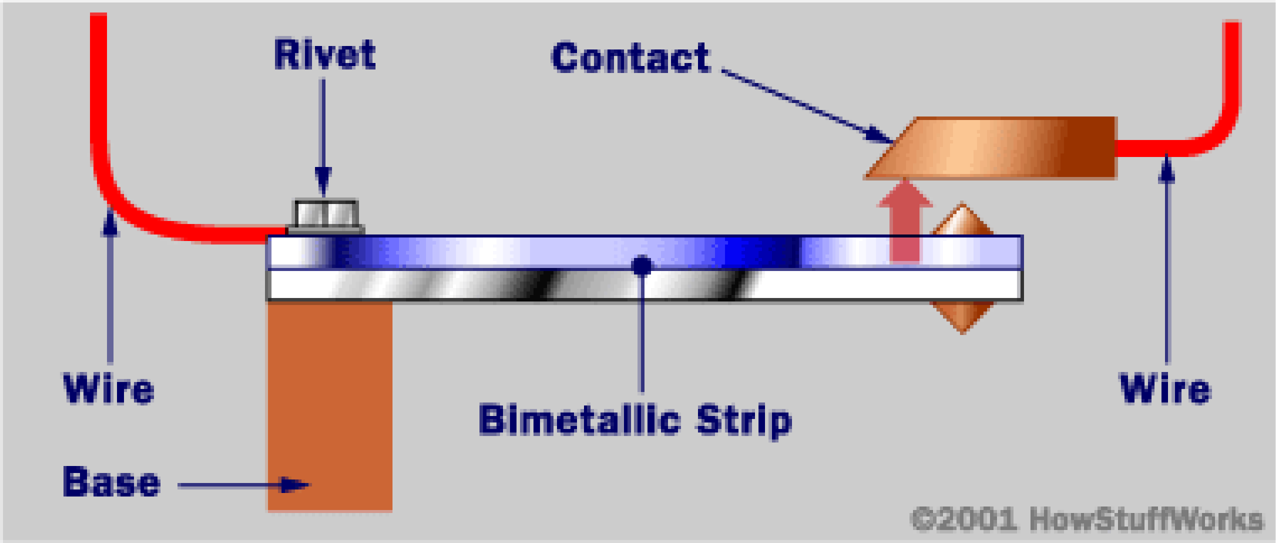
Applications of Thermal Expansion – Bimetallic Strip





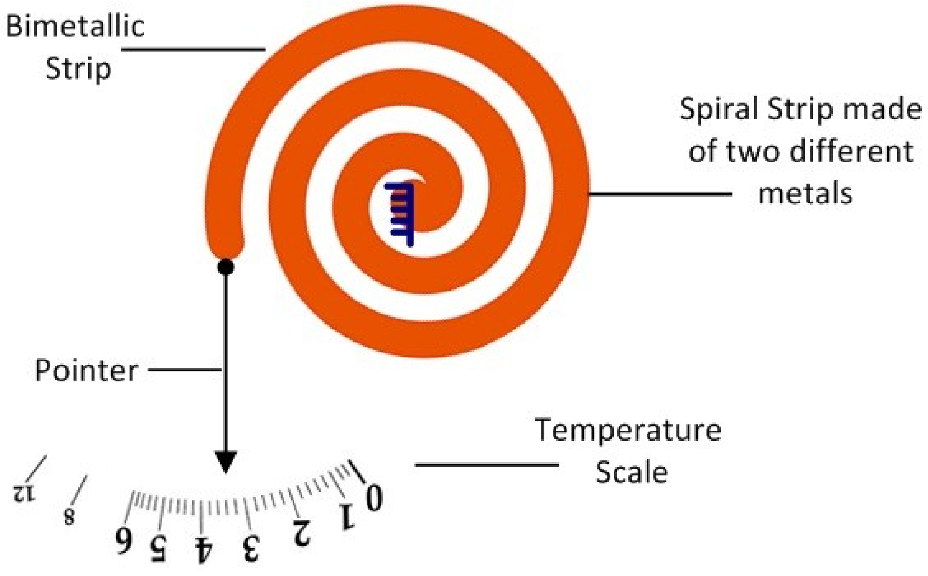
Bimetallic Strip

- Thermostats
  - Use a *bimetallic strip*
  - Two metals expand differently
    - Since they have different coefficients of expansion



Bi-metallic Strip

- Bimetallic strip Thermometer

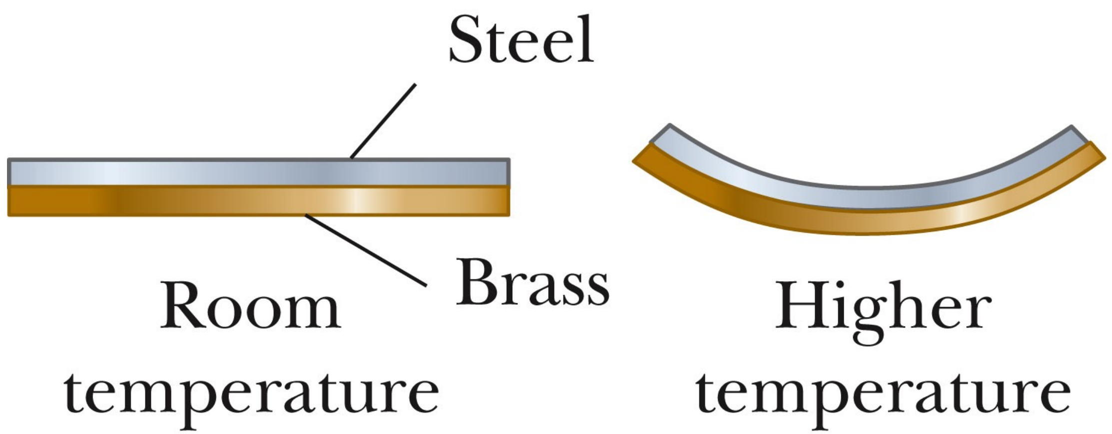


Spiral Strip Bimetallic Thermometer

Example

A railway track made of steel has a length of 20.0 m at 20°C. What will be the change in its length, and new length, when the temperature rises to 50°C during the day? Use $\alpha = 11 \times 10^{-6}\ C^{-1}$.

Applications of Thermal Expansion – Bimetallic Strip



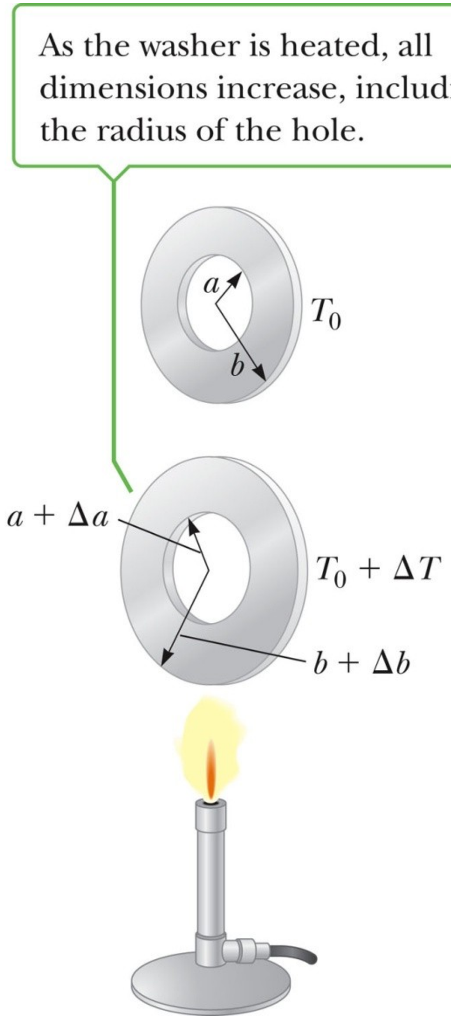
- Thermostats
  - Use a *bimetallic strip*
  - Two metals expand differently
    - Since they have different coefficients of expansion

Area Expansion

- Two dimensions expand according to

$$\Delta A = A - A_o$$
$$= \gamma A_o \Delta t,$$
$$\gamma = 2\alpha$$

  - γ is the coefficient of area expansion



Example

An aluminum Rectangle has size 2.00 x 5.00 m^2 at 0°C. What will be its area if its temperature increases to 90°C? Use $\alpha = 24 \times 10^{-6}\ C^{-1}$.

Using ΔL and ΔW:

Using Area equation:

Volume Expansion

- Three dimensions expand

$$\Delta V = \beta\, V_o\, \Delta t$$

$$\text{for solids, } \beta = 3\alpha$$

  - For liquids, the coefficient of volume expansion is given in the table

More Applications of Thermal Expansion

- Pyrex® Glass
  - Thermal stresses are smaller than for ordinary glass
- Sea levels
  - Warming the oceans will increase the volume of the oceans

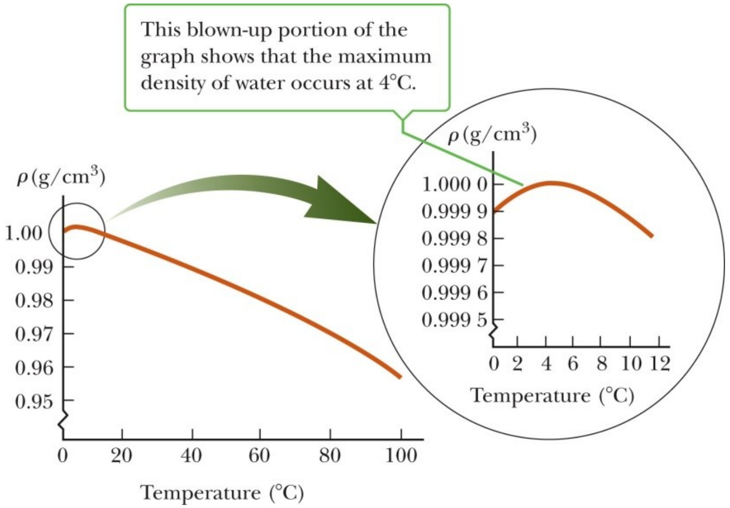
  $\beta = 2.07 \times 10^{-4}\ {}^\circ C^{-1}$

  Average depth = 4.0 km

Example

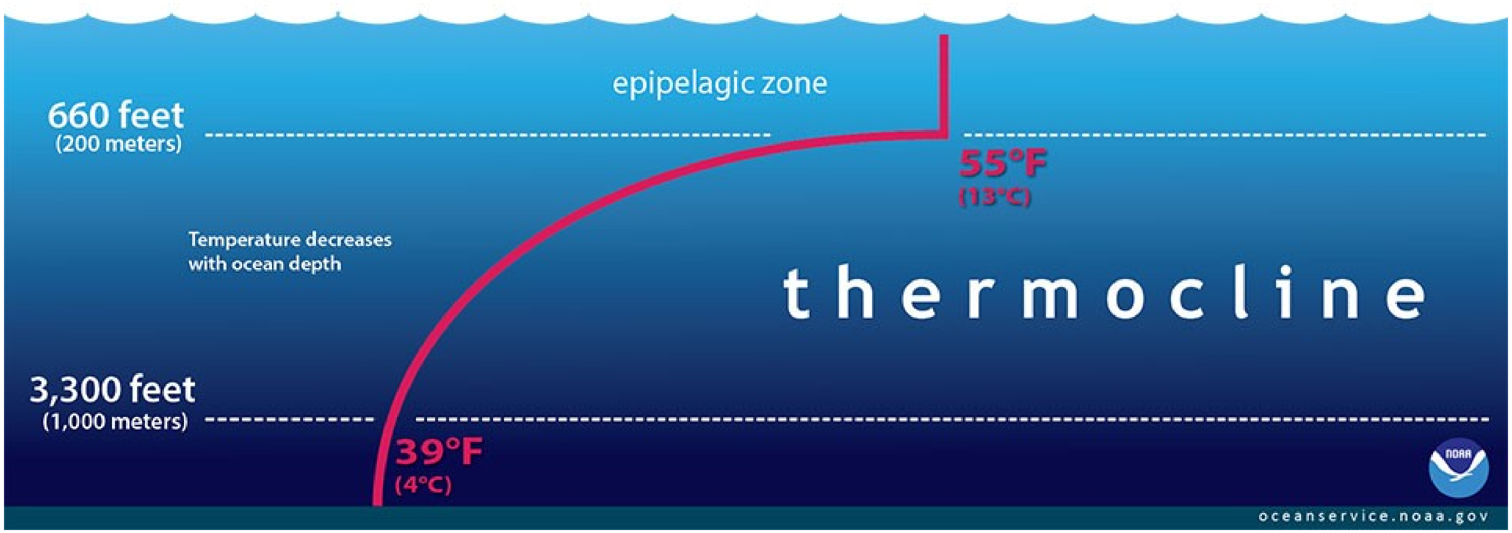
How much will sea level rise if the average water temperature increases by one degree celsius? Use $\beta = 2x10^{-4}\ C^{-1}$.

Unusual Behavior of Water



- As the temperature of water increases from 0°C to 4°C, it contracts and its density increases
- Above 4°C, water exhibits the expected expansion with increasing temperature
- Maximum density of water is 1000 kg/m^3 at 4°C

Temperature variation with depth



The Unusual Behavior of Water

Ideal Gas

- If a gas is placed in a container
  - It expands to fill the container uniformly
  - Its pressure will depend on the
    - Size of the container
    - The temperature
    - The amount of gas
- The pressure, volume, temperature and amount of gas are related to each other by an *equation of state*

Ideal Gas, cont

- The equation of state can be complicated
- It can be simplified if the gas is maintained at a low pressure
- Most gases at room temperature and pressure behave approximately as an ideal gas

Characteristics of an Ideal Gas

- Collection of atoms or molecules that move randomly
- Exert no long-range force on one another
- Each particle is individually point-like
  - Occupying a negligible volume

Moles

- It's convenient to express the amount of gas in a given volume in terms of the number of moles, n

$$n = \frac{\text{mass}}{\text{molar mass}}$$

- One mole is the amount of the substance that contains as many particles as there are atoms in 12 g of carbon-12

Avogadro's Number

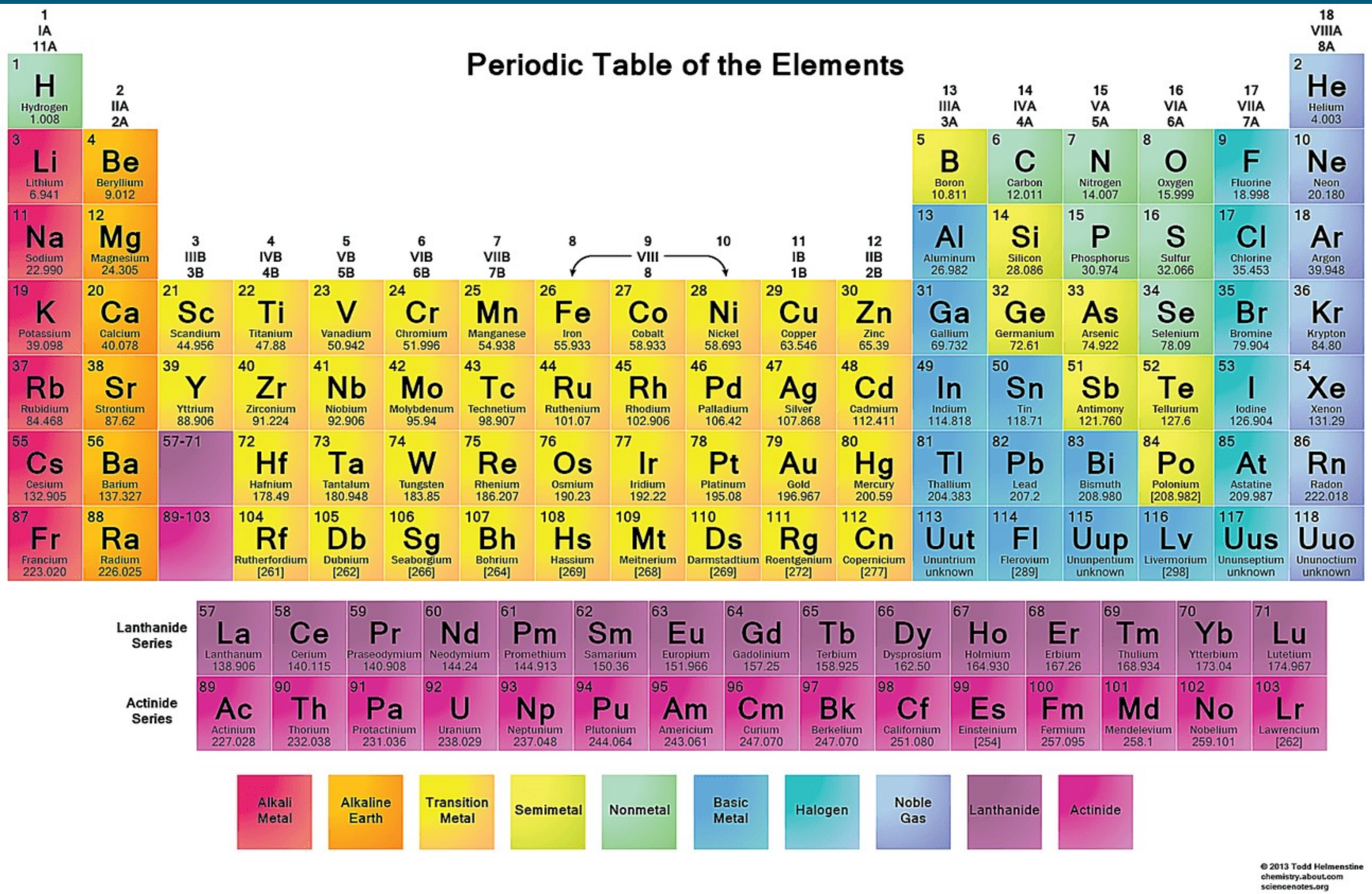
- The number of particles in a mole is called *Avogadro's Number*
  - N_A=6.02 x 10^{23} particles / mole
  - Defined so that 12 g of carbon contains N_A atoms
- The mass of an individual atom can be calculated:

$$m_{atom} = \frac{\text{molar mass}}{N_A}$$

Avogadro’s Number and Masses

- The mass in grams of one Avogadro's number of an element is numerically the same as the mass of one atom of the element, expressed in atomic mass units, u
- Carbon has a mass of 12 u
  - 12 g of carbon consists of N_A atoms of carbon
- Holds for molecules, also

Periodic table



Examples

- How many grams in one mole of:
- Ge
- H_2SO_4
- N
- N_2

How many atoms in 100 g of Carbon

How many atoms in 100 g of CO_2

Ideal Gas Law

Equation of state for an ideal gas: PV = n R T

- P = pressure
- V = volume
- n = number of moles of gas
- R is the *Universal Gas Constant*
- R = 8.31 J / mol·K or R = 0.0821 L·atm / mol·K
- T = Temperature (must be in kelvins)

Ideal Gas Law

PV = n R T

Consistent units:

n = moles = always a dimensionless number

T = Temperature = always in kelvin.

If: P = N/m^2 then V = m^3 and R = 8.31 J / mol·K

If: P = atm then V = liter and R = 0.0821 L· atm / mol·K

Example

- What will be the mass of air inside a tire, if its volume is 20 liter, pressure is 32 psig, and temperature is 40°C?

Ideal Gas Law, Alternative Version

- $P V = N k_B T$
  - k_B is *Boltzmann's Constant*
  - $k_B = R / N_A = 1.38 \times 10^{-23}$ J/ K
  - N is the total number of molecules
- $n = N / N_A$
  - n is the number of moles
  - N is the number of molecules

Ideal Gas Law, Alternative Version

- If the gas goes from some initial condition ‘1’ to some final condition ‘2’, and if the quantity (moles) of gas remains the same, then:

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Example

- A helium balloon at ground level has a volume of 5.00 m^3 where the pressure is 1 atm and temperature is 20°C. It is released, goes to a height where the pressure is 0.7 atm and temperature is -5°C. What will be the volume of the balloon at that height?

Kinetic Theory of Gases – Assumptions

- The number of molecules in the gas is large and the average separation between them is large compared to their dimensions
- The molecules obey Newton's laws of motion, but as a whole they move randomly
- The molecules interact only by short-range forces during elastic collisions
- The molecules make elastic collisions with each other and with the walls
- All molecules in the gas are identical
- https://en.wikipedia.org/wiki/Kinetic_theory_of_gases#/media/File:Translational_motion.gif

pressure

- Recall impulse: $I = F\,\Delta T = \Delta p$ (p is momentum)
- Then $F = \Delta p / \Delta T$ (i.e. change in momentum per unit time = $mv_f - mv_i$)
- If N molecules hit the wall per second, then $F = N\,\Delta p / \Delta T$
- $P = F/A$ (so pressure of the gas should depend on number of molecules, their masses and their speeds)
- If you pump in more molecules in the car's tire, its pressure increases.
- If you heat the tire, its pressure increases → speed increases.

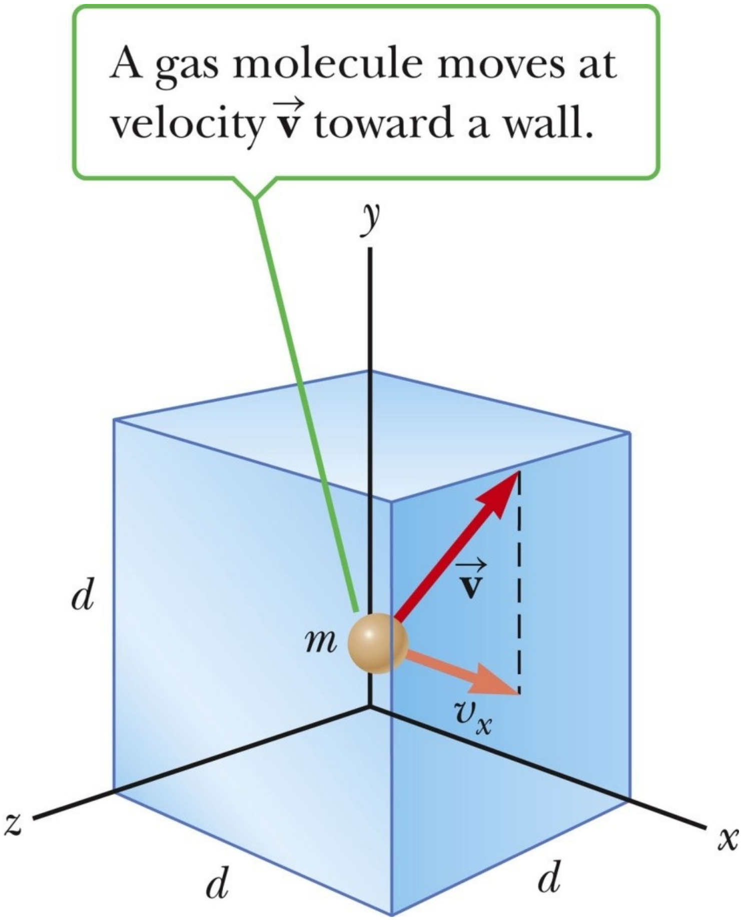
Pressure of an Ideal Gas

- The pressure is proportional to the number of molecules per unit volume and to the average translational kinetic energy of a molecule

$$P = \frac{3}{2}\left(\frac{N}{V}\right)\left(\frac{1}{2}m\overline{v^2}\right)$$

- Compare with

$$P\,V = N\,k_B\,T$$



Pressure, cont

- The pressure is proportional to the number of molecules per unit volume and to the average translational kinetic energy of the molecule
- Pressure can be increased by
  - Increasing the number of molecules per unit volume in the container
  - Increasing the average translational kinetic energy of the molecules
    - Increasing the temperature of the gas

Molecular Interpretation of Temperature

- Temperature is proportional to the average kinetic energy of the molecules
  - Temperature is a direct measure of the average molecular kinetic energy of the gas

$$\frac{1}{2}m\overline{v^2} = \frac{3}{2}k_B T$$

- The total translational kinetic energy is proportional to the absolute temperature

$$KE_{total} = \frac{3}{2}nRT$$

Internal Energy

- In a monatomic gas, the KE is the only type of energy the molecules can have

$$U = \frac{3}{2}nRT$$

- U is the *internal energy* of the gas
- In a polyatomic gas, additional possibilities for contributions to the internal energy are rotational and vibrational energy in the molecules

Speed of the Molecules

- Expressed as the *root-mean-square* (rms) speed

$$v_{rms} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3RT}{M}}$$

- At a given temperature, lighter molecules move faster, on average, than heavier ones
  - Lighter molecules can more easily reach escape speed from the earth

H_2 at 300 K:

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{2.0\times 10^{-3}\ \text{kg/mol}}} = 1.9\times 10^{3}\ \text{m/s}$$

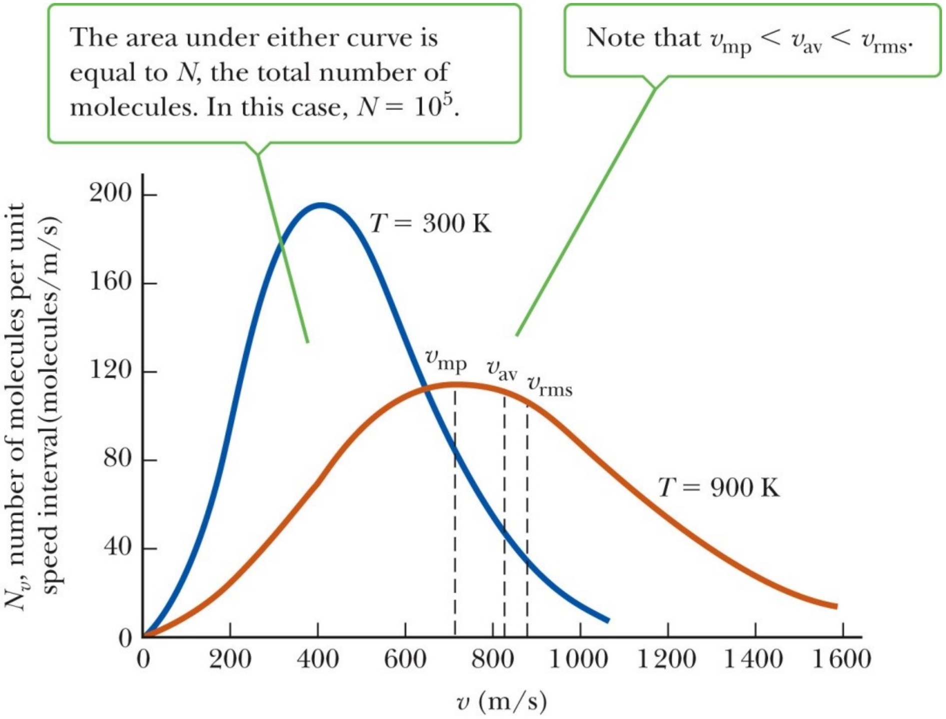
Some rms Speeds

Table 10.2 Some rms Speeds

Gas	Molar Mass (kg/mol)	v_{rms} at 20°C (m/s)
H_2	2.02×10^{-3}	1 902
He	4.0×10^{-3}	1 352
H_2O	18×10^{-3}	637
Ne	20.2×10^{-3}	602
N_2 and CO	28.0×10^{-3}	511
NO	30.0×10^{-3}	494
O_2	32.0×10^{-3}	478
CO_2	44.0×10^{-3}	408
SO_2	64.1×10^{-3}	338

Maxwell Distribution

- A system of gas at a given temperature will exhibit a variety of speeds
- Three speeds are of interest:
  - Most probable
  - Average
  - rms



Maxwell Distribution, cont

- For every gas, $v_{mp} < v_{av} < v_{rms}$
- As the temperature rises, these three speeds shift to the right
- The total area under the curve on the graph equals the total number of molecules